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Machine learning for stroke prediction using imbalanced data

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The research focused on predicting strokes, a significant threat to health and well-being. The primary challenge addressed was the use of a highly imbalanced dataset. Various data preprocessing techniques were employed to tackle this, enabling the construction and comparison of machine-learning models for stroke prediction. Among the models assessed, the random forest model proved to be the most effective, achieving precision, recall, and F1-score levels of 90%, along with an accuracy of 90%. A random forest classifier was also trained using optimal hyperparameters obtained via grid search on balanced data to highlight the limitations of relying solely on accuracy in classification tasks. This approach demonstrated the model's high accuracy of 96%, underscoring the impracticality of using accuracy as the sole metric for performance evaluation in imbalanced datasets. In conclusion, the research underscores the critical role of advanced data processing and machine learning techniques in enhancing stroke prediction. The successful application of the random forest model and the recognition of accuracy's limitations provide a robust framework for future studies and practical implementations in healthcare settings. This work advances the predictive capabilities for stroke and paves the way for improved approaches to healthcare strategies, ultimately aiming to save lives and enhance the quality of life for individuals at risk.

Keywords K-neighbors, Ada boost, Stroke, Random under-sampling, Random forest, Decision tree, Stacking, Machine learning, Grid search

Stroke is a leading cause of mortality and disability worldwide, accounting for approximately 1 in 21 deaths, with a person dying from stroke every 3 min and 17 s[1]. Global stroke mortality varies, with the highest rates observed in Central and Southeast Asia, Oceania, and sub-Saharan Africa. The financial burden of stroke is substantial, as patients often require long-term care, rehabilitation, and medical support, leading to financial instability for families[2].

Managing stroke patients is complex due to the diversity of symptoms, recovery rates, and the need for personalized treatment plans[3]. Effective monitoring requires regular check-ups, imaging, and assessments, placing a significant strain on healthcare resources. Additionally, stroke survivors frequently experience psychological challenges, including depression and anxiety, necessitating mental health support[4]. Predictive tools for early stroke risk detection can enhance preventive care, reducing severe outcomes through lifestyle modifications, medication, and close monitoring of high-risk individuals[5].

Despite being a preventable condition, many individuals remain unaware of their stroke risk, delaying medical intervention[6]. Machine learning (ML) offers a powerful approach to stroke prediction by analyzing patient data and identifying risk patterns[7]. However, the challenge lies in optimizing ML models for accurate stroke risk assessment, mainly when dealing with imbalanced datasets.

This study focuses on preprocessing techniques to enhance data quality, developing and optimizing ML models, and evaluating their performance using multiple metrics[8,9]. Various ML classifiers, including Decision Tree, Random Forest, K-Nearest Neighbors, AdaBoost, and Stacking, are examined to determine the most effective approach. Special attention is given to handling data imbalance, as stroke datasets typically contain far more non-stroke cases, affecting model reliability.

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Prior research has explored different ML models for stroke prediction, with ensemble learning methods like Random Forest and Support Vector Machines achieving high accuracy[10–14]. However, many studies overlook key factors such as dataset selection, feature importance, and data preprocessing, which impact model effectiveness.

This study presents a novel approach to stroke prediction by addressing data imbalance and optimizing hyperparameters for improved classification. By comparing multiple ML models and assessing their precision, recall, and F1-score, we provide a comprehensive evaluation framework. Our findings contribute to healthcare decision-making by enhancing early stroke detection, enabling timely intervention, and ultimately improving patient outcomes.

Materials and methods

Bibliometric analysis of machine learning methods for stroke prediction

We utilized VOSviewer[15], a software tool for constructing and visualizing bibliometric networks, to conduct a comprehensive literature review on machine learning methods for predicting strokes. VOSviewer facilitates the creation of maps using data exported from Scopus. In our analysis, we examined 20,000 Scopus papers using the keywords "machine learning" and "stroke prediction." These articles were categorized into four clusters.

The first cluster, represented by the map of the item "artificial intelligence" (Fig. 1), includes 108 links, 521 total link strength, and 934 occurrences. This cluster highlights the significant role of artificial intelligence in stroke prediction research.

The second cluster, depicted in the map of the item "neural network" (Fig. 2), comprises 73 links, 127 total link strength, and 341 occurrences. This cluster emphasizes the importance of neural networks in developing predictive models for stroke.

The third cluster, shown in the map of the item "brain stroke" (Fig. 3), contains 9 links, 11 total link strength, and 24 occurrences. This cluster focuses on the specific application of machine learning techniques to brain stroke prediction.

Finally, the fourth cluster, illustrated in the map of the item "stroke prediction" (Fig. 4), includes 20 links, 34 total link strength, and 96 occurrences. This cluster underscores the direct efforts in predicting stroke occurrences using machine learning methods.

In conclusion, the bibliometric analysis conducted with VOSviewer provided important insights into machine learning for stroke prediction. We categorized 20,000 Scopus papers into four key clusters: artificial intelligence, neural networks, brain stroke, and stroke prediction, showcasing the major contributions and focal points within this research area. This analysis underscores the critical role of machine learning in enhancing stroke prediction and creates opportunities for further investigation and refinement of predictive models.

Comparative analysis of machine learning models for stroke prediction

In this section, we conduct a comparative analysis of various machine learning models for stroke prediction. By examining the most relevant existing studies, we aim to identify the strengths and limitations of different approaches. Table 1 provides a detailed comparison of the tools and tasks reviewed in these studies, highlighting the diverse methodologies and their respective outcomes. This analysis will help inform future research endeavors by acknowledging the valuable insights and contributions of each study.

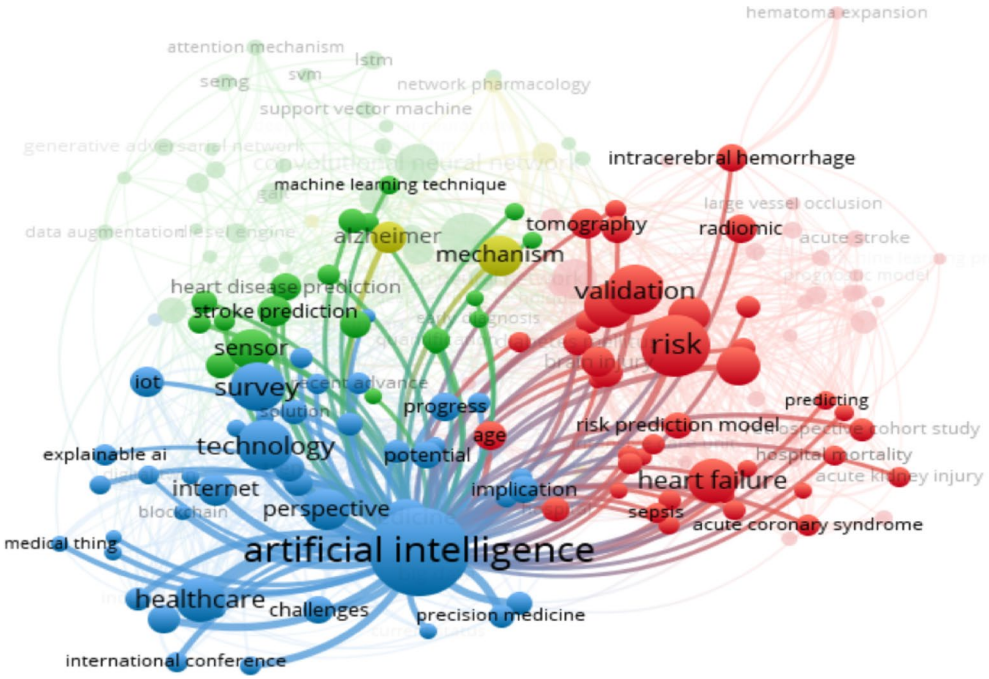


Fig. 1. The map of the item "artificial intelligence".

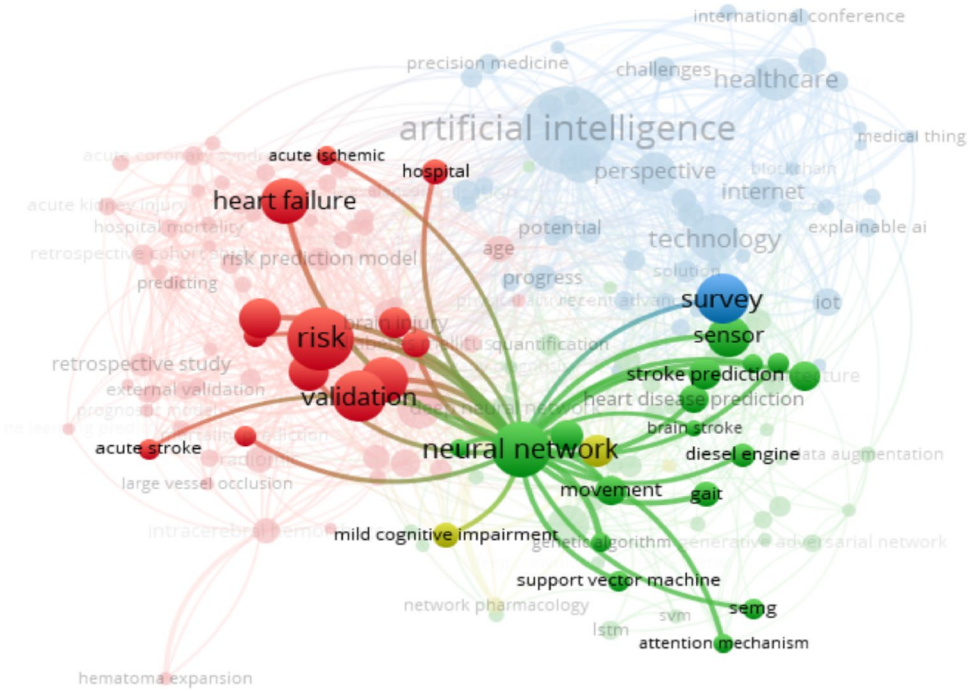


Fig. 2. The map of the item "neutral network".

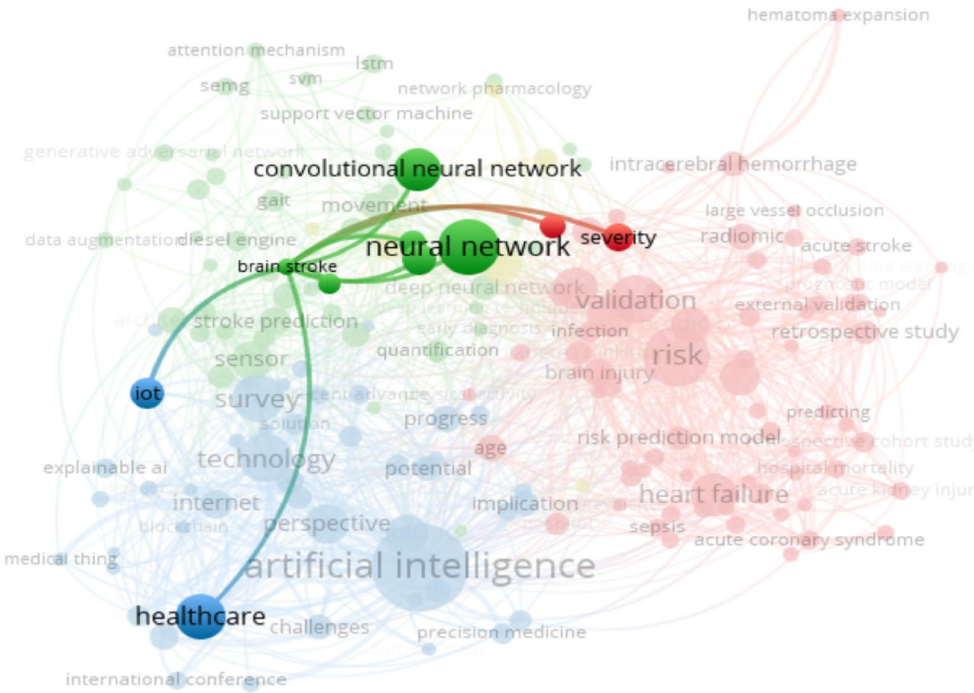


Fig. 3. The map of the item "brain stroke".

The application and comparison of various machine learning models—support vector machines (SVM), deep neural networks (DNN), random forest, decision trees, the naive Bayes algorithm, and logistic regression—revealed interesting results[14]. Among these, the Random Forest model achieved the highest accuracy at 96.01%, highlighting the effectiveness of ensemble methods in reducing errors in loss functions.

In another study, different machine-learning algorithms were tested on a medical dataset containing 3,178 patients[16]. Logistic regression, random forest, and XGBoost were applied, resulting in accuracy values of 0.89%,

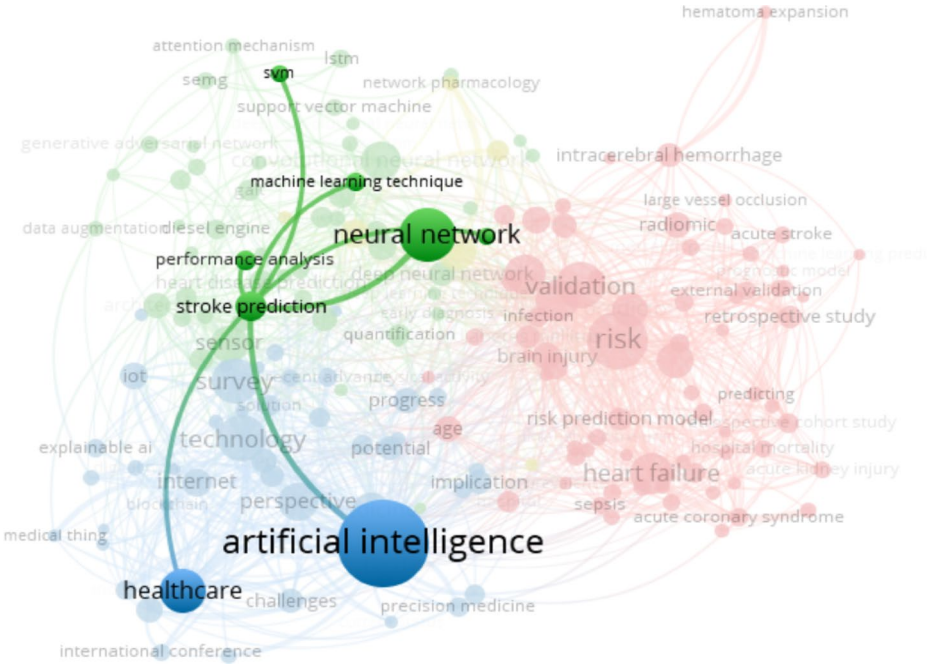


Fig. 4. The map of the item "stroke prediction".

0.89%, and 0.88%, respectively. However, it's important to note that this dataset, based on patient parameters collected in Japan, may not apply to other countries due to variations in patient characteristics.

A novel approach using a modified genetic folding algorithm based on various symptoms has been proposed for stroke prediction[17]. This method surpassed logistic regression by 4.2%, the naive Bayes method by 1.2%, decision trees by 17.2%, and the support vector machine by 3.2%, achieving an accuracy of 83.2%. Nevertheless, the small dataset size of only about 1,000 patients limits the accuracy and effectiveness of the model.

Another stroke prediction model employs a DNN trained on private data, achieving an impressive accuracy of 98.17%[18]. However, this model also faces limitations due to the small sample size of only 180 patients, which can lead to unreliable results and a lack of comparison with other models.

A machine learning algorithm for stroke prediction using an SVM classifier trained on private data has achieved an accuracy of 91.04%[19]. This study's limitation lies in the dataset's few examples of strokes, potentially introducing bias and reducing the prediction accuracy for other data.

Researchers using Naive Bayes and Decision Tree algorithms to identify individuals at risk of stroke achieved an accuracy of 94%. However, this study[20] was limited by insufficient information about the data's origin and preparation process, and it did not compare its results with similar studies, which restricts the reliability of its conclusions.

Another proposed model based on supervised learning for stroke prediction reached an accuracy of 94.6%[21]. The study's reliability is potentially compromised by a lack of detailed information about data selection and preparation methods.

A comprehensive analysis used six interpretable algorithms and four feature selection strategies to develop 131 stroke prediction models[22]. Although 21 models achieved an accuracy above 0.89, and 110 models exceeded an accuracy of 0.91, the study was limited by a small sample size of 170 patients and a lack of cross-validation to verify prediction accuracy.

Researchers have also developed a CNN model for predicting early-stage stroke probability, achieving an accuracy of 95.5%. However, the study[23] did not consider data processing methods, which might affect the results' reliability.

A performance comparison of stroke prediction models using various supervised machine learning algorithms was conducted to select the optimal algorithm. Although the Random Forest model achieved the highest accuracy at 92.24%, the study[24] lacked detailed descriptions of data processing methods and parameter selection, making it difficult to verify the results.

Various machine learning models, including logistic regression, decision tree classification, random forest classification, and a voting classifier, have been developed to predict the probability of stroke using physiological parameters. Despite Random Forest demonstrating the highest performance with approximately 96% accuracy[25], the study did not provide detailed descriptions of model parameters and data pre-processing methods.

Finally, several machine learning models have been evaluated for predicting stroke risk, with the ensemble method showing high efficiency, as indicated by AUC and accuracy metrics. This approach achieved an AUC of 98.9% and an accuracy of 97.4%. However, the study[26] did not compare the effectiveness of this method with other stroke risk prediction methods, which limits the broader applicability of the findings.

No	Title	Tools	Task	Year	Reference
1	A comparative analysis of machine learning classifiers for stroke prediction: A predictive analytics approach	Machine support vector, random forest, K-nearest neighbor, decision tree, naive Bayes, voting classifier, AdaBoost, gradient boosting, multi-layer perceptron, nearest centroid	Comparison of the effectiveness of various machine learning classification methods for predicting stroke risk	2022	13
2	A machine learning ensemble classifier for prediction of brain strokes	Random forest, K-nearest neighbor, logistic regression, machine support vector, naive Bayes	Development of an ensemble of machine learning classifiers for predicting stroke risk	2022	14
3	Analyzing the performance of stroke prediction using ML	Logistic regression, decision tree, random forest, K-nearest neighbor	Analysis of the effectiveness of classification algorithms	2021	16
4	Ensemble-learning based framework for brain stroke prediction	Machine support vector, DNN, random forest, decision tree, naive Bayes, logistic regression	Development of a framework based on an ensemble of machine learning classifiers for predicting stroke risk	2021	17
5	Development of of machine learning models to predict probabilities and types of stroke at prehospital stage: the Japan urgent stroke triage score using machine learning (JUST-ML)	Logistic regression, random forest, XGBoost	Development of machine learning models to predict the probability and type of stroke at the prehospital stage	2022	18
6	Genetic folding (GF) algorithm with minimal Kernel operators to predict stroke patients	Genetic folding (GF) algorithm	Application of the genetic folding algorithm with minimal kernel operators for predicting stroke in patients	2022	19
7	Brain stroke prediction model using deep neural network (DNN)	DNN (Deep neural network)	Development of a stroke risk prediction model based on deep neural networks	2021	20
8	An artificial intelligence-based bio-medical stroke prediction and analytical system using a machine learning approach	Support vector machine	Development of an artificial intelligence biomedical system for stroke prediction using machine learning approaches	2022	21
9	ML approach for brain stroke prediction using Ist database	Naive Bayes, decision tree	Application of machine learning methods for predicting stroke risk based on the Indian Stroke Technology (IST) database	2022	22
10	Optimizing predictions of brain stroke using machine learning	Support vector machine, random forest, K-nearest neighbor	Improving stroke prediction using machine learning methods	2022	23
11	Prediction of long-term stroke recurrence using machine learning models	Logistic regression, extreme gradient boosting, gradient boosting machine, random forest, support vector machine, decision tree	Long-term stroke recurrence risk prediction using machine learning methods	2021	24
12	Prediction of stroke disease using deep CNN based approach	CNN	Development of a deep learning model based on convolutional neural networks for stroke disease prediction	2022	25
13	Predictive model analysis of stroke disease based on machine learning	Decision tree, random forest, Gaussian Bayes	Development of a predictive model for stroke disease based on machine learning methods	2023	26
14	Stroke disease detection and prediction using robust learning approaches	Logistic regression, decision tree classification, random forest classification, voting classifier	Development of a model for stroke prediction and detection based on robust machine learning method	2021	27
15	Stroke risk prediction with machine learning techniques	Stacking	Development of a stroke risk prediction model using machine learning methods	2022	28

Table 1. Comparative analysis of the tools and task-reviewed resources.

The review and analysis of the selected articles revealed a significant interest in the topic, yet many studies exhibit several common issues:

- Some studies overly complicate models by combining numerous machine-learning methods[27,28] and artificial intelligence techniques[29,30], increasing computational resources without improving accuracy.
- Despite achieving relatively high accuracy levels, many studies fail to adequately address the challenge of unbalanced data processing.
- There's often a lack of optimization in parameter selection for different machine learning methods.

Novelty and value

The paper presents a novel approach to stroke risk assessment using machine learning. It addresses the challenge of unbalanced datasets in stroke prediction, where there are significantly more non-stroke cases than stroke cases, which affects the model's accuracy. The study uses various data preprocessing methods to address this imbalance and improve the reliability and accuracy of the model.

It evaluates several machine learning models, including Decision Tree, Random Forest, K-Nearest Neighbors, AdaBoost, and Stacking classifiers. The study compares these models and highlights the superior performance of the Random Forest model in terms of precision, recall, F1 score, and overall accuracy. The research also uses a grid search method for hyperparameter optimization to fine-tune the Random Forest classifier for optimal performance.

The paper highlights the limitations of using accuracy alone to evaluate performance in unbalanced datasets and advocates for a more comprehensive evaluation approach. The implications of the study are significant for healthcare practitioners, as improved predictive models can aid in the early identification of individuals at risk for stroke, potentially reducing severe stroke outcomes.

Advanced data processing techniques, such as random under sampling and other balancing methods, ensure that both stroke and non-stroke cases are accurately represented in the models. The comprehensive analysis of performance metrics provides valuable insights into the strengths and weaknesses of each approach, helping to understand the impact of data imbalances on model performance and guiding future research in this area.

The paper also highlights the importance of using multiple performance metrics beyond accuracy in imbalanced data scenarios. It provides a framework for healthcare practitioners to identify high-risk individuals early, enabling better preventive measures and improved patient outcomes.

Key steps in processing and analyzing medical data for stroke prediction

We outline crucial steps for processing medical data to predict strokes. Timely diagnoses are vital for prevention, yet many high-risk individuals remain unaware of their condition, delaying medical assistance. Machine learning techniques can enhance early stroke risk detection, enabling prompt preventative actions and better patient outcomes. Our research evaluates various machine learning methods for diagnosing stroke risk, aiming to identify the most effective approaches to improve predictive accuracy and support early intervention strategies.

One of the key factors contributing to the occurrence of strokes (Fig. 5) is the lack of timely diagnosis. Often, individuals with a high risk of stroke are unaware of their condition and, therefore, do not seek medical assistance. Furthermore, even when they seek medical attention, the diagnostic process can be complex, causing delays in essential treatment and potentially leading to serious consequences.

Utilizing machine learning methods can significantly enhance the early identification of patients at high risk of stroke and recommend necessary preventive measures. This approach increases the chances of successful treatment, improving patients' health and quality of life. The research problem is to investigate the effectiveness of various machine learning methods for accurately predicting the risk of stroke. The goal is to identify the most effective techniques to enhance predictive accuracy and improve early intervention strategies.

Justification of the selected machine learning models

We will now examine the characteristics of our dataset, which underpins the stroke risk prediction job. The dataset was obtained from the DataHack Analytics firm Vidhya[31]. The dataset consists of:

- 11 columns representing various indicators and parameters of patients.
- 4981 rows containing the values of these indicators and parameters for each patient.

Data preprocessing is crucial in tackling real-world challenges such as data incompleteness, inconsistency, lack of trends, and errors in today's data landscape.

Real-world data often exhibits inconsistencies, incompleteness, or missing values in certain samples or trends, leading to potential errors in its quality. Data preprocessing offers an effective approach to address these issues.

This process aims to prepare the initial data for subsequent analysis and processing by executing operations and transformations. These operations enhance data quality, making it more suitable for training machine learning models and facilitating better interpretation and comprehension.

Key steps involved in data preprocessing include outlier removal, missing value imputation, categorical feature encoding, data scaling, and various other operations. These steps are essential for preventing model overfitting[31].

In our scenario, considering the structure and content of our data and the objectives outlined in this paper, we will execute the following data preprocessing operations:

- Outlier Handling: Identifying and addressing outliers in the dataset.
- Categorical Data Encoding: Converting categorical variables into numerical representations suitable for machine learning algorithms.
- Handling Unbalanced Data: Implementing techniques to address class imbalance in the dataset.
- Dataset Splitting: Dividing the dataset into training and testing sets for model evaluation.
- Attribute Scaling: Normalizing or standardizing the numerical attributes to a common scale.

Regression models are utilized to predict numerical values (attributes). In contrast, classification models are employed to solve problems where it's necessary to determine the category or class to which an object belongs.

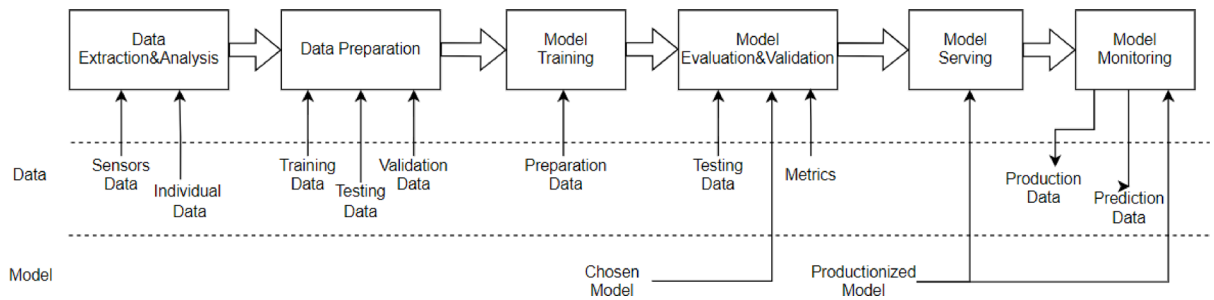


Fig. 5. Model of processing & analyzing the medical data to predict stroke occurrence.

Given our task involves predicting whether a patient will have a stroke, with only two possible outcomes (yes or no), regression models are not well-suited for this type of problem.

Thus, the most appropriate solution for our task is to utilize a machine-learning classification model. Having determined the model type needed, we can select specific classification models for our task.

One such model is the Decision Tree algorithm, commonly used for classification tasks. It operates by constructing a tree-like structure where each internal node represents a "question" based on features that help define an object's class. By traversing the tree and answering these questions, the model determines the class to which each object belongs.

The decision tree model offers several advantages[32], including its ease of interpretation for humans, compatibility with both qualitative and quantitative information, and its independence from prior data normalization.

The decision tree model has notable disadvantages. It is susceptible to overfitting, particularly if the tree is excessively deep, and it is also vulnerable to noise and outliers in the data, which can impact its performance and accuracy.

Random Forest is a machine learning algorithm for classification tasks in the ensemble methods category. The fundamental concept of the Random Forest classification algorithm involves constructing a robust classifier by amalgamating the outcomes of numerous decision trees, each built on random subsets of both data and features.

The primary advantages of the Random Forest model lie in its ability to yield often exceptionally high classification accuracy owing to the aggregation of multiple decision trees. Additionally, the random selection of data subsets and features for each tree aids in mitigating overfitting. A notable drawback of this model is its increased complexity in interpreting results compared to a single decision tree, as it can be challenging to discern which tree made a particular decision and why.

K-Neighbors is another machine learning algorithm suitable for classification tasks using the K-Nearest Neighbors (K-NN) method[33]. The fundamental principle behind this classification algorithm is assigning a class to a new object based on the classes of its "nearest neighbors" from a known dataset.

The primary advantages of the K-Neighbors model include its interpretability, as it relies on the simple notion that similar objects belong to similar classes, and its consideration of the data context, as classification is based on the "similarity" between objects.

Nevertheless, a key disadvantage of the K-Neighbors model lies in its computational complexity, particularly with large datasets, as it necessitates calculating distances to all objects in the training set for each new feature. Additionally, selecting the optimal value of K is crucial, as it significantly impacts classification quality.

AdaBoost is a machine learning algorithm categorized under ensemble methods that are commonly used for classification tasks. It aims to enhance classification accuracy by combining multiple weak classifiers, such as decision trees, into a single strong classifier.

The primary advantages of the AdaBoost classifier include its ability to significantly improve classification accuracy by leveraging basic classifiers that may not be effective individually. Additionally, the model adapts to the data by assigning more weight to incorrectly classified objects in previous iterations.

However, a notable disadvantage of AdaBoost is its susceptibility to noise and outliers in the data. Because the algorithm focuses on objects misclassified in previous training stages, it may become less efficient when dealing with noisy or outlier-rich datasets. This can lead to overfitting, where the ensemble adjusts too much to individual anomalies, thus hindering its ability to generalize on new data.

Stacking, another ensemble method used for classification problems, differs from traditional methods like Random Forest or AdaBoost. The fundamental concept of Stacking involves combining predictions from multiple base classifiers or models to improve classification quality.

The main advantages of the Stacking classifier lie in its ability to enhance classification accuracy by integrating predictions from diverse models. It excels in handling complex classification problems that require amalgamating different types of information for better results.

On the downside, the Stacking classifier entails complexity in configuration and imposes a high computational cost for model calculation and training. Selecting appropriate base models and configuring the meta-model can be challenging, as well as determining the optimal number and types of base models.

To address our problem, we will employ five classification models: Decision Tree classifier, Random Forest classifier, K-Neighbors classifier, AdaBoost classifier, and Stacking classifier.

Procedures for selecting the optimal parameters of the models

In this section, we discuss the procedures for selecting the optimal parameters of machine learning models, focusing on the importance of hyperparameters in fine-tuning model performance. Hyperparameters play a crucial role[33], as they are key factors in fine-tuning machine learning models. Hyperparameters in machine learning are model parameters that are not automatically learned during the training process but need to be set before training begins. They influence the learning process and the structure of the model.

Hyperparameters define the conditions and customize the learning process, determining how the model learns and adapts to the training data.

When dealing with a classification problem, the choice of hyperparameter search method depends on various factors, such as the size of the dataset, computational resources, and time constraints. Given that our dataset has 4981 rows and 11 columns, which is not very large, the Grid Search method seems suitable.

Grid Search is a hyperparameter tuning method where all possible combinations of hyperparameters predefined by the user are tested. This systematic approach allows for evaluating different combinations of hyperparameters and identifying the ones that yield the best results on a validation dataset.

Grid Search provides a structured approach to evaluating different hyperparameter combinations, ensuring the selection of the best configuration for optimal model performance. Its main advantages include ease of implementation, as it simplifies the process of determining the most effective hyperparameter values. It is particularly efficient for small datasets, such as the one used in this study (4,981 rows and 11 columns), allowing for a comprehensive exploration of multiple parameter settings within a reasonable time. Additionally, its systematic nature ensures no possible combination is overlooked, leading to a thorough and reliable hyperparameter optimization process.

A potential drawback arises when dealing with many hyperparameters and numerous possible values. In such cases, the computational cost of Grid Search may become prohibitive.

Analyzing the results of the chosen Hyperparameters for the Selected Machine Learning Models was presented. Understanding and optimizing hyperparameters is crucial for maximizing the performance of machine learning models. In this section, we outline the main findings and critical factors for each chosen model:

For the Decision Tree Classifier, key hyperparameters include:

The Decision Tree Classifier determines split quality using the criterion parameter, where "gini" is the default option, though "entropy" is also available. Both methods perform well, with the choice depending on dataset characteristics and computational efficiency. The splitter parameter defines the strategy for selecting splits, with "best" optimizing accuracy and "random" promoting diversity, which can be beneficial in ensemble learning. To control complexity and prevent overfitting, the max_depth parameter limits tree growth, with values typically set between 10 and 100 to maintain a balance between predictive power and generalization.

For the Random Forest Classifier, key hyperparameters include:

For the Random Forest Classifier, key hyperparameters include N_estimators, Criterion, Max_depth, and Max_features. N_estimators, which defaults to 100 but is often set between 100 and 1000, controls the number of trees and can improve accuracy through ensemble averaging, albeit with higher computational costs. The Criterion parameter, typically "gini" or alternatively "entropy," determines how splits are evaluated, balancing efficiency and accuracy based on dataset characteristics. Max_depth limits the growth of trees—defaulting to unlimited—but setting it between 10 and 50 can prevent overfitting and enhance generalization. Finally, Max_features defines the number of features considered for each split, where a higher value may improve splits yet also increase computation and risk reducing tree diversity.

For the K-Neighbors Classifier, important hyperparameters include:

The K-Neighbors Classifier requires careful tuning of several hyperparameters to optimize its performance. We know that tha number of neighbors (n_neighbors), typically set to 5 with a common range from 3 to 10, determines how many surrounding points influence the classification, where lower values can make the model overly sensitive to noise and higher values may smooth the decision boundary. The weights parameter, by default set to "uniform" to treat all neighbors equally, can be switched to "distance" so that closer neighbors have a more significant influence, potentially improving accuracy when neighbor relevance decreases with distance. Additionally, the algorithm parameter, which defaults to "auto," allows the system to select the most efficient method (such as ball_tree, kd_tree, or brute force) based on the dataset's size and dimensionality. In contrast, the metric parameter is generally set to "Minkowski"—a flexible option that encompasses both Euclidean and Manhattan distances—though it can be adjusted to other metrics depending on the specific characteristics of the data.

For the Ada Boost Classifier, key hyperparameters include:

For the AdaBoost classifier, the default configuration employs a DecisionTreeClassifier with a maximum depth of 1 (a decision stump) as the base estimator, which helps to mitigate overfitting by sequentially correcting errors made by previous classifiers. The ensemble's size is controlled by the n_estimators parameter, which defaults to 50; while increasing this value can reduce bias and variance, it also raises computational costs, so values typically range between 50 and 500 depending on the dataset's complexity. Additionally, the learning_rate parameter, set by default at 1.0, determines the influence of each classifier's contribution to the final output. Lowering the learning rate can enhance accuracy through a more gradual training process, albeit at the expense of longer training times, with practical values often falling between 0.01 and 1.0.

For the Stacking Classifier, key hyperparameters include:

The Stacking Classifier utilizes a collection of base classifiers, defined in the estimators parameter, where different classifiers are chosen to increase diversity and capture various patterns within the data. This diversity helps improve the model's performance by balancing out the unique weaknesses of individual classifiers. The final_estimator usually, a simple classifier like logistic regression, combines the predictions of the base classifiers to make the final decision, acting as a meta-learner. The CV parameter controls the number of splits for cross-validation during base classifier training, with the default 5-fold setting being a good balance between computational cost and model robustness. Increasing CV folds can enhance validation reliability but requires more computational resources.

In a Stacking Classifier, multiple base classifiers are combined to enhance prediction performance. The model first constructs a diverse set of estimators by selecting different classifiers, each identified by a unique name, to capture various patterns in the data and balance individual errors. These base classifiers feed into a final estimator—often a simple linear model such as logistic regression—that serves as a meta-learner, integrating the diverse outputs into a more reliable prediction. Additionally, cross-validation (typically 5-fold) is employed during the training of these base classifiers to ensure consistent performance across data subsets. However, adjusting the number of validation blocks can further refine model tuning at the cost of increased computational effort.

Each model has its unique set of hyperparameters that can significantly influence performance. Understanding and optimizing these parameters are crucial in building effective machine-learning models tailored to specific datasets and tasks.

Model performance indicators

This section explores the performance indicators used to assess machine learning models for stroke prediction. Assessing the accuracy of a prediction is a nuanced endeavor, as there's no one-size-fits-all metric. Practical experimentation is often required to determine a specific scenario's most suitable Key Performance Indicator (KPI). Each performance metric has strengths and weaknesses, proving effective in certain contexts while falling short in others. Metrics for evaluating the efficacy of machine learning models in classification tasks serve as crucial benchmarks in model development and refinement. They enable us to gauge how well a model fulfills its intended task and facilitate comparisons between different models and classification methodologies.

Evaluation metrics play a crucial role in assessing machine learning model performance, each offering distinct advantages depending on the dataset and classification task.

Accuracy provides an easy-to-interpret measure of how often the model correctly classifies instances. It is particularly effective when the dataset has a balanced class distribution, as it reflects overall correctness without being skewed by class imbalances.

Precision and recall are essential when dealing with imbalanced data, as they help determine the model's ability to identify a specific class correctly. These metrics provide deeper insights into both the accuracy and completeness of predictions, making them particularly useful in applications where false positives or false negatives have significant consequences.

The F1 is especially useful when class distributions are uneven, ensuring a more comprehensive assessment of model performance in such scenarios.

Another critical evaluation tool is the confusion matrix, which allows for a more granular analysis of model weaknesses. This helps identify specific areas for improvement and optimize decision-making strategies based on observed errors.

In our model evaluation, we will prioritize F1 Score, Precision, and Recall as primary metrics. The Confusion Matrix will serve as a supplementary metric, while Accuracy will be used to compare with the main metrics to address the imbalance in our dataset.

Model evaluation involves checking its performance and generalization capability using a test dataset containing unseen instances. Metrics such as accuracy, precision, recall, and the F1 score can be used to quantitatively assess the classifier's success in predicting the correct classes for each sample. Accuracy (1) is the primary criterion for many classification tasks and determines the proportion of correctly classified samples in the test dataset[9,10].

These equations provide a comprehensive understanding of the model's performance in predicting the correct classes for each sample and help identify areas for improvement in classification tasks.

Results

In the dataset, we encounter various types of data in machine learning:

Categorical Features:

- Nominal: gender, ever_married, work_type, residence_type, smoking_status.
- Binary: hypertension, heart_disease, stroke.
- Numerical Features:
- Continuous: avg_glucose_level, bmi.
- Discrete: age.

In the context of data visualization, we should commence by analyzing the distribution of these features.

From Fig. 6, several observations can be made:

- The dataset contains more women than men, with a difference of approximately 850–900 individuals.
- The number of individuals without a history of heart disease significantly surpasses those with a history of heart disease by roughly 4.5 times.
- Individuals without hypertension outnumber those with hypertension by about 4 times.

The most prevalent occupation type is private work, followed by self-employment. In contrast, the number of individuals engaged in public work is nearly equivalent to the number of unemployed individuals. Regarding residential areas, approximately equal numbers of individuals reside in urban and rural areas.

Analyzing Fig. 7 yields the following conclusions:

Age Distribution:

- The age distribution across the dataset appears relatively even, indicating a balanced representation of different age groups.
- There is a slight increase in the number of individuals between the ages of 40 and 70 compared to other age ranges.
- The age groups of 10–15 years and 85 years exhibit the lowest frequencies within the dataset.
- Average Glucose Level Distribution:
- Most individuals in the dataset have average glucose levels ranging from 85 to 90, indicating a common range.
- Conversely, average glucose levels of 175–180 and 250 + are less frequent, suggesting a lower occurrence.
- There is a notable decline in the frequency of individuals with average glucose levels exceeding 100.

These observations provide insights into the dataset's age and average glucose level distribution (Fig. 8), highlighting common trends and variations across different age groups and glucose level ranges.

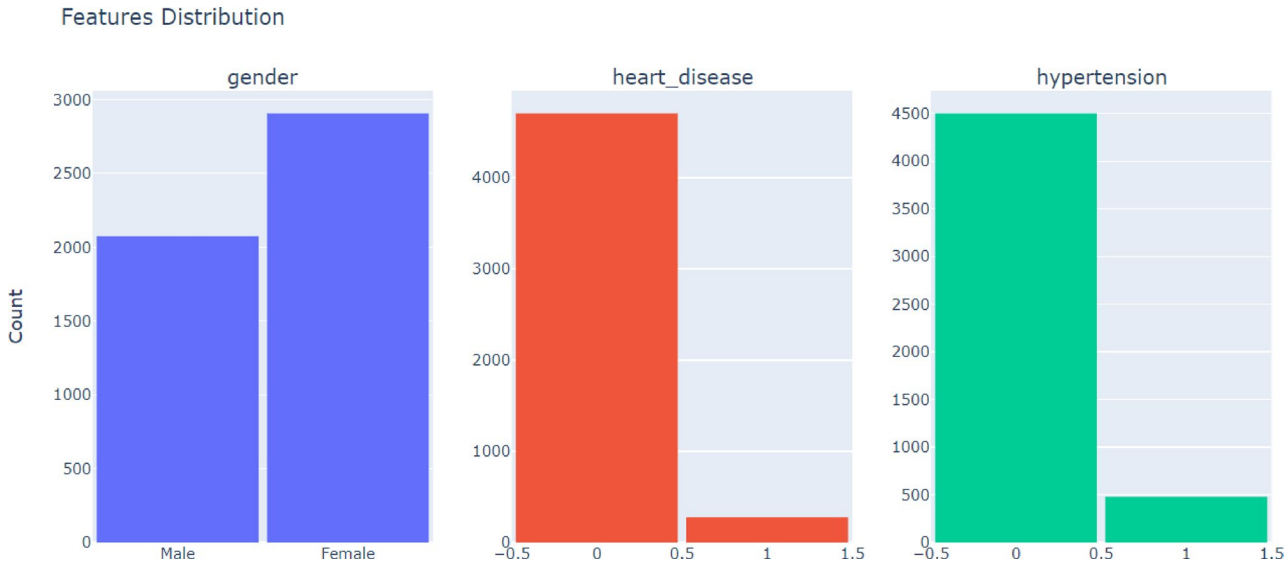


Fig. 6. Analysis of feature distributions - gender, heart disease, and hypertension.

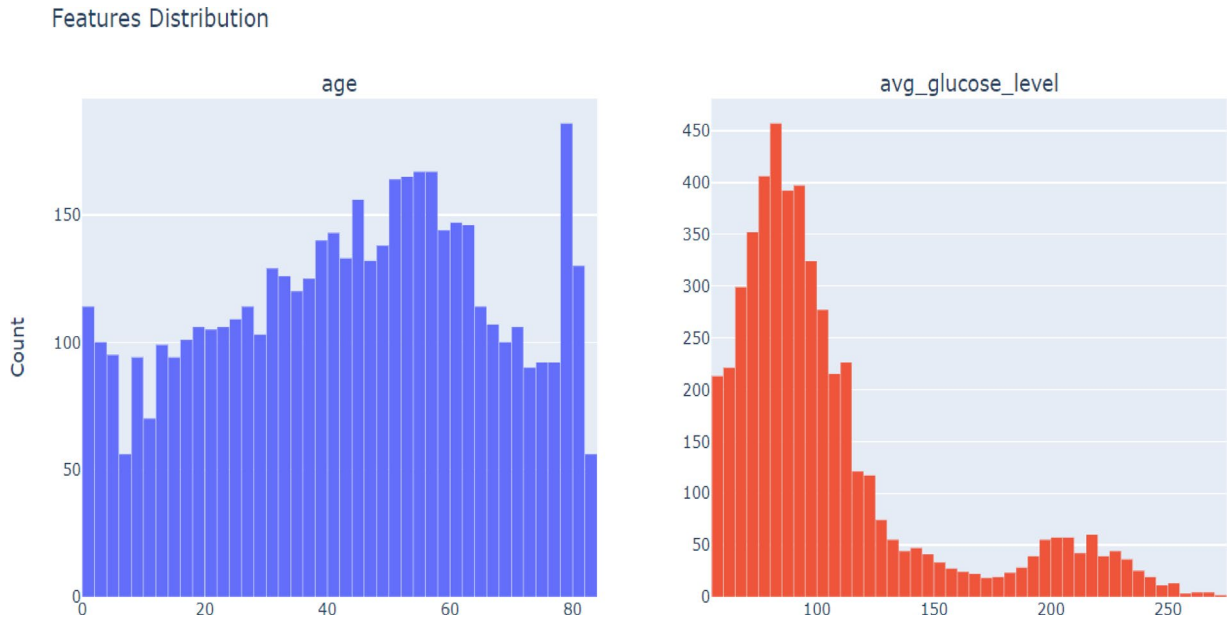


Fig. 7. Analyzing the distribution of age and average glucose level.

Body Mass Index (BMI) Distribution:

- The distribution of BMI in the dataset follows a normal distribution pattern.
- Patients with a BMI ranging from 23 to 33, corresponding to the median of the normal distribution, are most commonly observed.
- BMI values between 0 and 10 and between 45 and 60 occur with similar frequencies, further indicating the normal distribution of this trait.
- Smoking Status Distribution:
- The majority of individuals in the dataset are categorized as never having smoked, followed closely by those with unknown smoking status, which lacks information about their smoking habits.
- The number of individuals categorized as former smokers and current smokers is approximately equal.
- While individuals categorized as smokers represent the smallest group, they still constitute a significant portion of the dataset.

These findings shed light on the distribution of BMI and smoking status within the dataset, providing valuable insights into the prevalence of different BMI ranges and smoking habits among the individuals included.

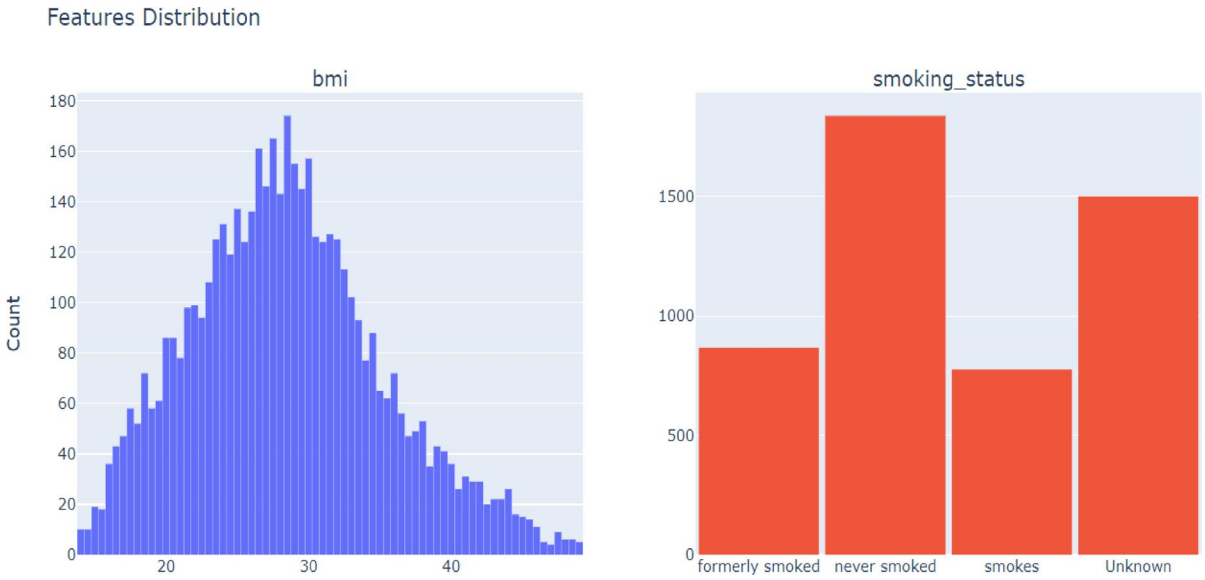


Fig. 8. Analyzing reveals the following insights.

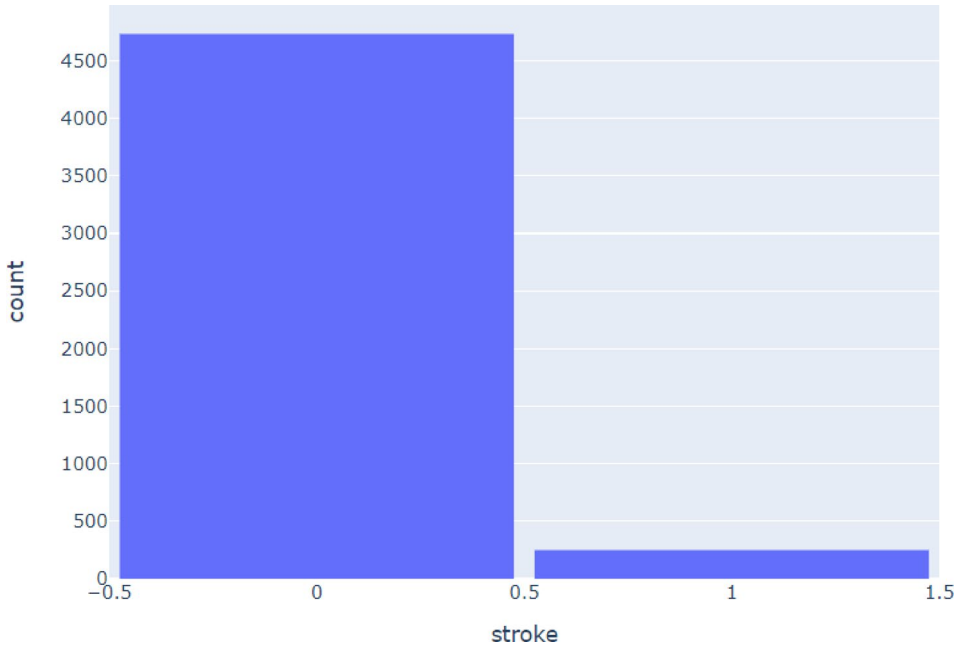


Fig. 9. Distribution of the trait – stroke.

Additionally, it emphasizes the importance of analyzing the distribution of the target variable, "stroke," to assess data balance and inform predictive modeling (Fig. 9).

From the analysis, several conclusions emerge:

Data Imbalance:

- The dataset exhibits significant imbalance, with a vast majority of individuals having no history of stroke.
- The proportion of individuals without a stroke vastly exceeds those who have experienced one.
- The ratio of individuals without a stroke to those with one is approximately 9:1.

Importance of Data Balancing:

- Balancing the dataset is crucial to ensure that models yield high-quality results.

Addressing the data imbalance is imperative to prevent models from skewed toward predicting the majority class and facilitate accurate predictions for both classes (Table 2).

The analysis yields the following conclusions regarding the dataset:

Age:

- The maximum age recorded in the dataset is 82 years.
- The average age of individuals in the dataset is 43 years.

Glucose Levels:

- The maximum recorded glucose level is 271.74.
- The average glucose level among individuals is 105.94.
- The minimum recorded glucose level is 55.12.

Body Mass Index (BMI):

- The maximum BMI observed is 48.9.
- The average BMI among individuals is 28.50.
- The minimum BMI recorded is 14.

These statistics highlight a notable variance and presence of outliers, particularly in characteristics such as BMI and average glucose levels.

Next, we'll analyse the distribution of our features based on whether individuals have experienced a stroke. See Figs. 10, 11, 12, 13, 14 and 15 for details.

Below are the conclusions derived from the analysis:

- While the distribution of stroke occurrence is relatively similar regardless of gender, there is a slightly higher likelihood of men having a stroke compared to women.
- The overall proportion of individuals who have experienced a stroke remains low for both genders.

The conclusions drawn from the analysis are as follows:

- Stroke occurrences are predominantly observed in individuals aged 40 and above.
- While strokes have been recorded at ages as young as 15 and 32, they are more commonly observed in older age groups.
- The distribution of stroke occurrences between 40 and 65 follows a normal pattern.
- The stroke occurrences between the ages of 65 and 82 also exhibit a normal distribution.
- The age at which strokes most commonly occur is around 80 years old.

The conclusions that follow are as follows:

- Individuals who have had hypertension and stroke constitute a notably higher percentage compared to those who have not had hypertension but have experienced a stroke.
- The proportion of stroke occurrences among individuals with hypertension is relatively small.
- The proportion of stroke occurrences among those without hypertension is very minimal.
- Hypertension seems to have a relatively modest impact on the likelihood of experiencing a stroke.

The following conclusions can be made:

- Individuals who have experienced heart problems and stroke constitute a significantly higher percentage compared to those who have not had heart problems but have experienced a stroke.
- The proportion of stroke occurrences among individuals with heart problems is relatively small.
- The proportion of stroke occurrences among those without heart problems is minimal.
- The relationship between heart disease and stroke variables seems relatively weak.

The following observations can be made:

	Age	avg_glucose_level	bmi
Count	4981	4981	4981
Mean	43.42	105.94	28.50
Std	22.66	45.08	6.79
Min	0.08	55.12	14
25%	25	77.23	23.7
50%	45	91.85	28.1
75%	61	113.86	32.6
Max	82	271.74	48.9

Table 2. Statistical characteristics of numerical characteristics.

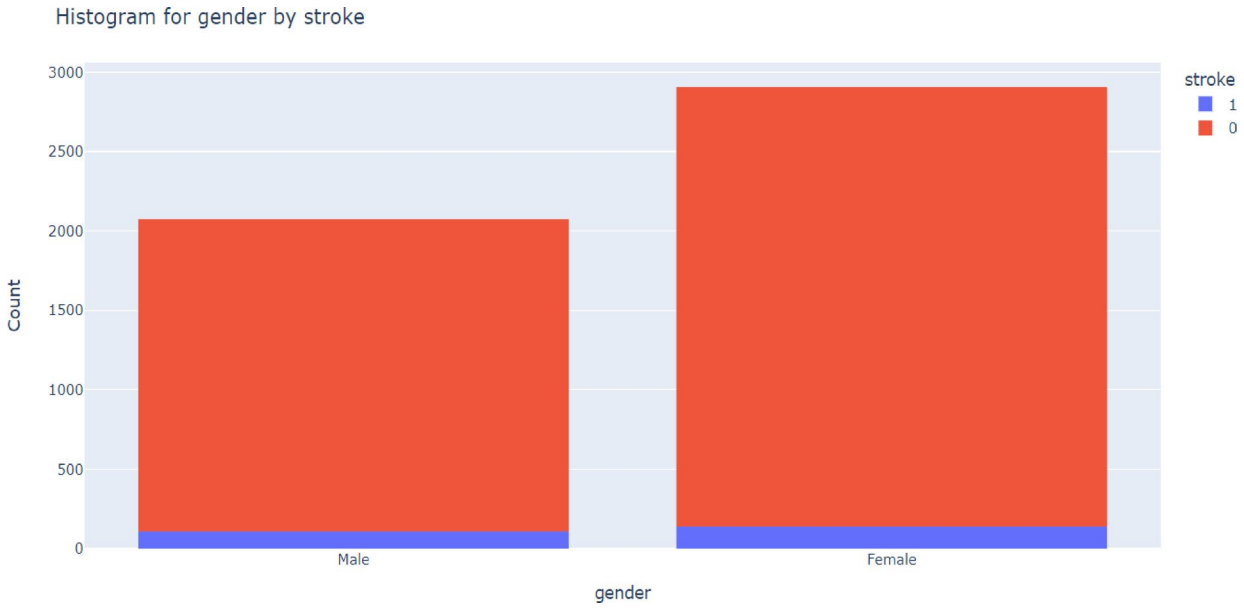


Fig. 10. Distribution of the gender trait depending on the stroke.

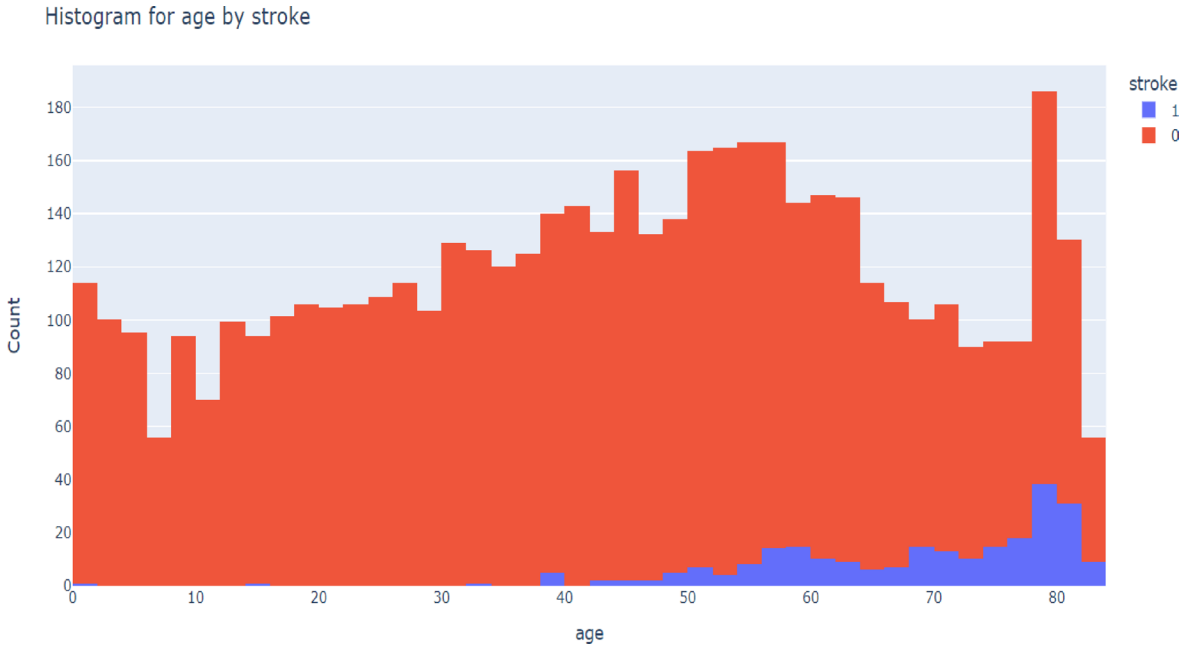


Fig. 11. The distribution of the age trait depends on the stroke.

- The distribution of stroke occurrences among individuals with a body mass index (BMI) between 25 and 30 exhibits a pattern resembling a normal distribution.
- Stroke occurrences are evident in individuals with a BMI of 17 or higher.
- The highest incidences of stroke are observed among individuals with BMI values of 23, 30, and 32.
- Conversely, stroke incidences are least frequent among individuals with BMI values of 17, 18, 48, and 49.

The following conclusions can be inferred:

- The proportion of individuals who have experienced a stroke and either smoked in the past or currently smoke is approximately the same, at around 12%.
- The proportion of individuals who have never smoked but have experienced a stroke is relatively low.
- Smoking does no.3t appear to have a very strong impact on the occurrence of stroke.

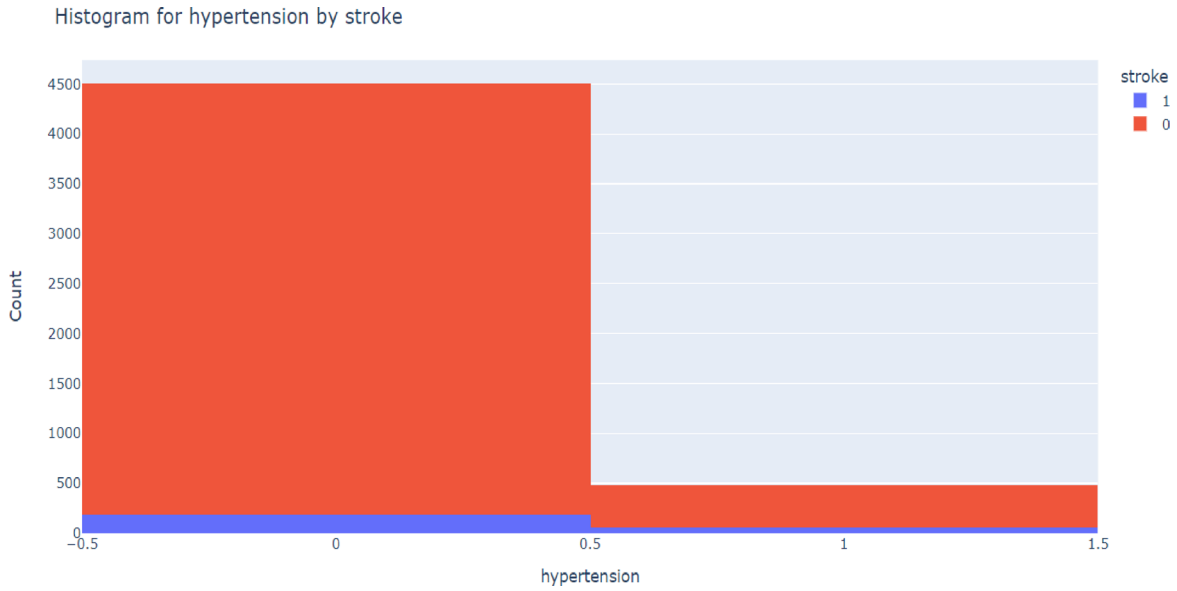


Fig. 12. The distribution of the hypertension trait depends on the stroke.

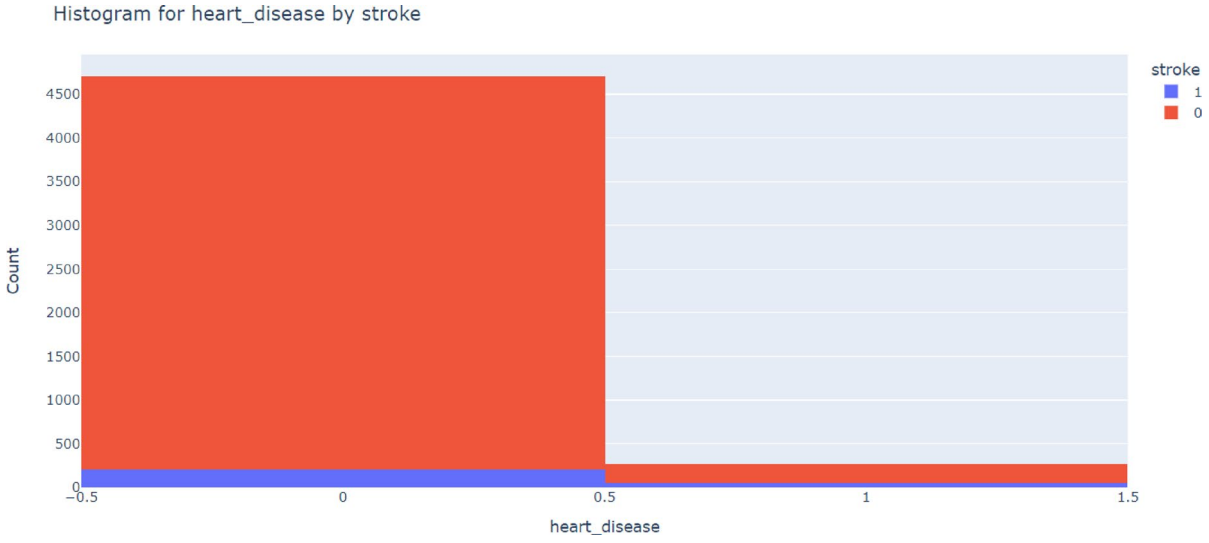


Fig. 13. The distribution of the heart_disease trait depends on the stroke.

Subsequently, we will find outliers inside our dataset (Fig. 16). We will start by analyzing the characteristics of outliers.

We deduced that outliers are present in our dataset, particularly in the following attributes: average glucose level and body mass index (BMI).

The scatter plot depicts the relationship between average glucose level and BMI. See Fig. 17 for details.

The presence of anomalies in the data is evident in this graph. As previously stated, we will exclude them using the interquartile range method. We will now proceed to incorporate the categorical characteristics in our dataset. We will first encode information that may be expressed in binary form, like gender, marital status, and kind of dwelling. In this encoding process, we will replace the following values:

- For gender: 'male' becomes 1, 'female' becomes 0.
- For ever_married: 'Yes' becomes 1, 'No' becomes 0.
- For residence_type: 'urban' becomes 1, 'rural' becomes 0.

The outcomes of this operation are illustrated in Fig. 18.

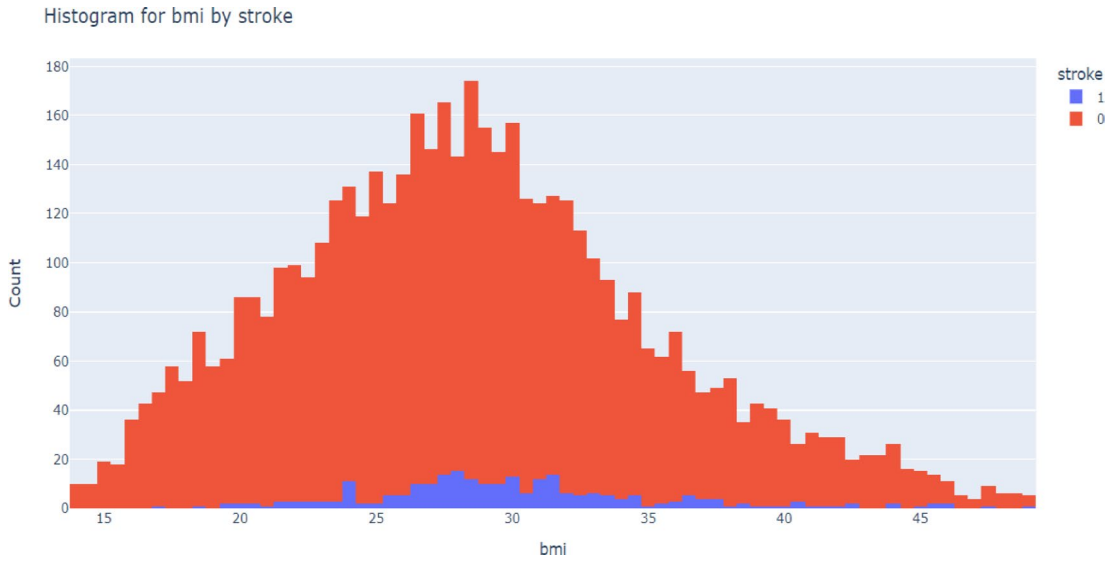


Fig. 14. The distribution of the BMI trait depends on the stroke.

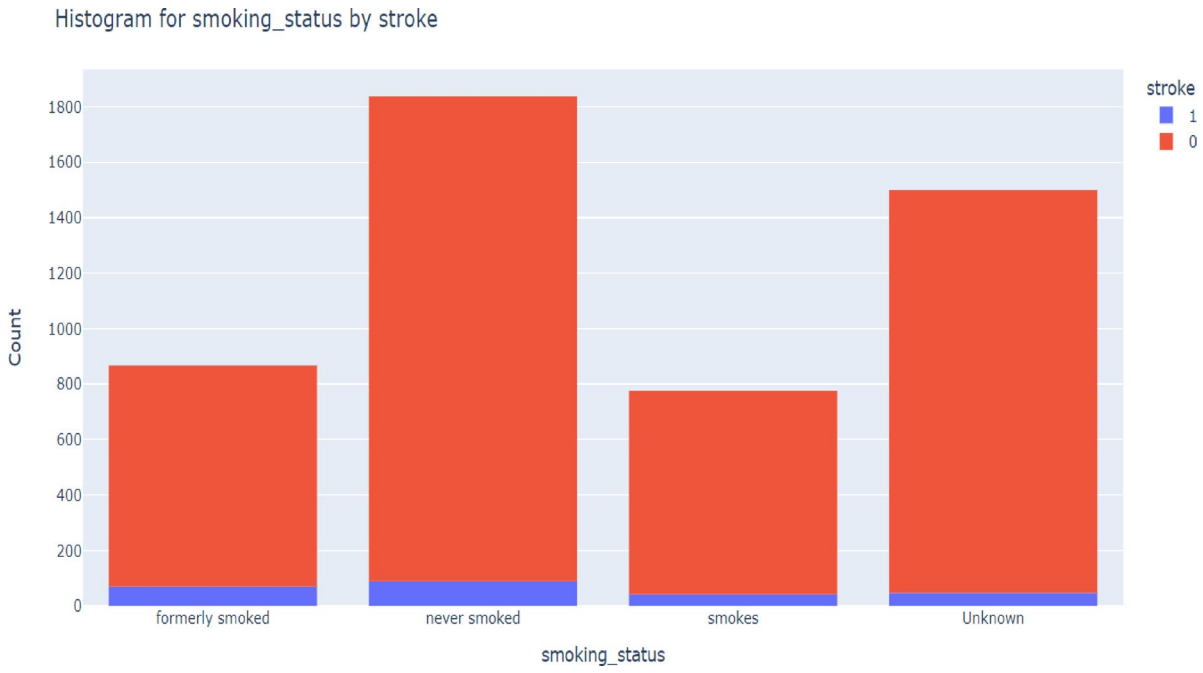


Fig. 15. The distribution of the smoking status trait depends on the stroke.

Upon completing all data preparation tasks (except data partitioning), we will analyze the data correlation matrix to discern possible dependencies prior to the development of our machine learning models. Figure 19 illustrates the correlation matrix:

The following observations can be made from this figure:

- The likelihood of experiencing a stroke is most strongly correlated with the characteristic of age.
- Marital status also shows a moderate correlation with the likelihood of having a stroke, albeit weaker than age.
- There is a weak correlation between the probability of having a stroke and the presence of hypertension and heart problems.
- Other characteristics appear to have minimal to no impact on the likelihood of experiencing a stroke.

In conclusion, we will divide the data into training and testing sets for modeling purposes. We will use an 80 – 20 split, where 80% of the data is allocated for training and 20% for testing. The model will be trained on 5,679 rows of data and assessed with 1,420 rows of data.



Fig. 16. Box graph of numerical features.

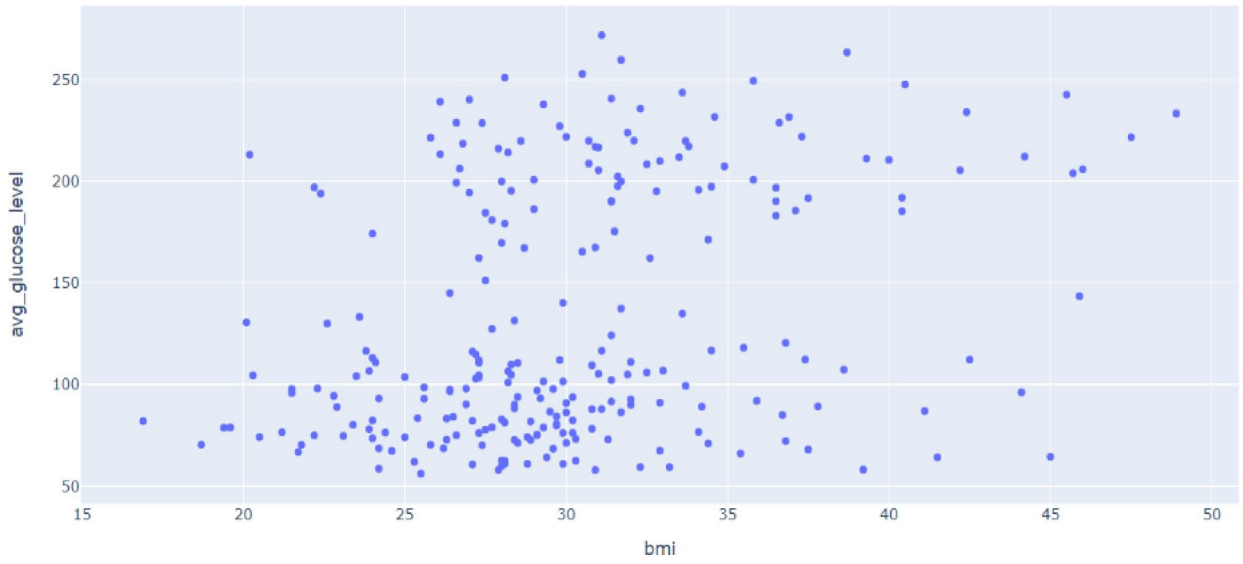


Fig. 17. Scatter graph avg_glucose_level and BMI.

	age	hypertension	heart_disease	work_type	avg_glucose_level	bmi	smoking_status	stroke	gender	ever_married	Urban
0	67.0	0	1	Private	105.943562	36.6	formerly smoked	1	1	1	1
1	80.0	0	1	Private	105.920000	32.5	never smoked	1	1	1	0
2	49.0	0	0	Private	105.943562	34.4	smokes	1	0	1	1
3	79.0	1	0	Self-employed	105.943562	24.0	never smoked	1	0	1	0
4	81.0	0	0	Private	105.943562	29.0	formerly smoked	1	1	1	1

Fig. 18. Trait coding results – gender, ever_married and Residence_type.

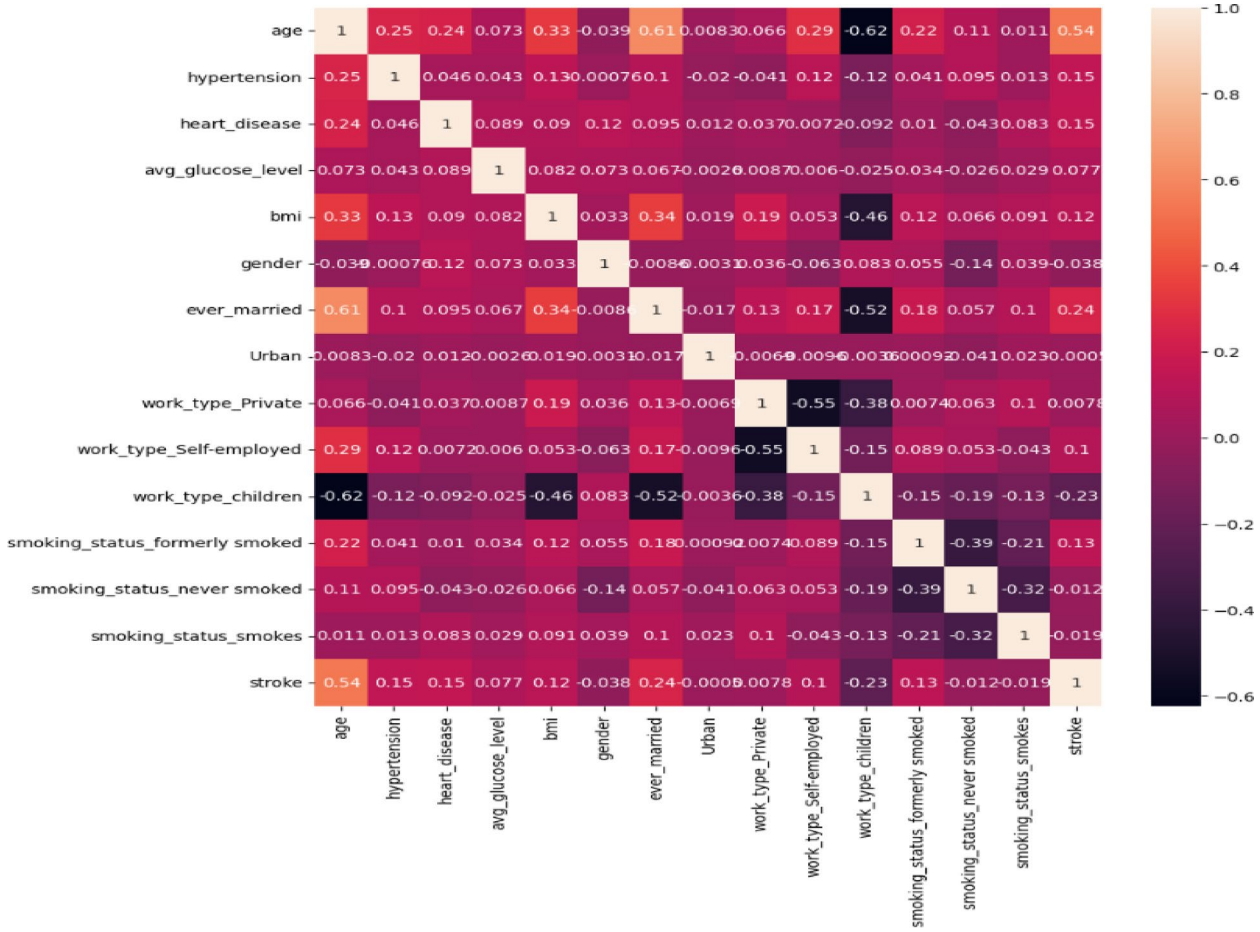


Fig. 19. Data correlation matrix in the data set.

Model	Precision	Recall	f1-score	Accuracy
Random forest classifier (improved)	0.93 for class 0 and 0.89 for class1	0.94 for class 0 and 0.86 for class 1	0.9	0.91
Random forest classifier (base)	0.93 for class 0 and 0.89 for class1	0.94 for class 0 and 0.86 for class 1	0.86	0.9
Stacking classifier (base)	0.94 for class 0 and 0.87 for class1	0.93 for class 0 and 0.88 for class 1	0.87	0.91

Table 3. Top model results.

Discussion

Next, we will examine the results of training and testing our models. Initially, we will assess the performance of the basic models with their default hyperparameters on the test data.

From these findings, we can draw the following conclusions:

- The F1-score has improved by 4% and now stands at 90%, indicating a high level of performance.
- The model has demonstrated balanced classification, with precision and recall almost identical for both classes.
- The number of misclassifications in class 0 (no stroke) remained at 69; class 1 (stroke) decreased to 52.

Subsequently, when creating an improved model using the acquired hyperparameters, similar results were achieved compared to the basic Stacking Classifier model. This outcome could be attributed to the model reaching its maximum potential on such a challenging dataset. In summary, it can be inferred that the basic models, namely the Random Forest Classifier and Stacking Classifier, along with the Random Forest Classifier model enhanced by the Grid Search method, yielded the best performance.

The outcomes of the top-performing models are outlined in Table 3.

From the table and the overall results obtained during the training of our models, several conclusions can be made:

Model	Class	Orecision	Recall	f1-score	Accuracy
Random forest classifier	0	0.93	0.94	0.86	0.9
	1	0.87	0.86		
Decision tree classifier	0	0.89	0.91	0.81	0.87
	1	0.82	0.80		
K-neighbors classifier	0	0.84	0.95	0.82	0.86
	1	0.92	0.75		
Ada boost classifier	0	0.82	0.86	0.7	0.79
	1	0.73	0.68		

Table 4. Comparative analysis of using classifiers without SMOTE method for data.

- The Random Forest Classifier model improved by the Grid Search method emerged as the best performer for our task. It showcased consistent results for both classes with an impressive F1-score of 90%, signifying robust performance.
- The Stacking Classifier model also demonstrated commendable performance, albeit slightly less stable than the improved Random Forest Classifier. Nevertheless, it achieved respectable results with an F1 score of 87%.
- Between the two basic models, the Random Forest Classifier outperformed the Stacking Classifier.

Consequently, the Random Forest model proved to be the most suitable option for our issue. Attaining an accuracy of almost 90% on the test data demonstrated exceptional competence. To enhance the comparison of findings and assess the influence of data set imbalance on model training, we will develop a Random Forest Classifier model using the hyperparameters that produced optimal outcomes without using the SMOTE technique (Table 4). We shall then analyze the results of this model.

These results show several conclusions regarding the model's performance trained on highly unbalanced data.

1. Poor Performance and Lack of Stability: The model exhibits poor performance and stability, particularly in classifying class 1 (stroke) instances. This indicates that the model struggles to generalize to minority class instances.
2. Failure to Classify Class 1 Instances: The model fails to classify any class 1 (stroke) instances correctly. This is a critical issue as the model's primary objective is to detect strokes.
3. Predominant Prediction of Class 0: The model predominantly predicts class 0 (no stroke), with only two misclassifications in this category. While achieving a high accuracy of 96%, the model's ability to correctly classify instances of class 1 is severely compromised.
4. Inadequacy of Accuracy Metric: Despite the high overall accuracy, this metric doesn't accurately reflect the model's actual performance. It emphasizes the dominance of the majority class (class 0) predictions, masking the model's inability to identify instances of class 1.
5. Poor Precision, Recall, and F1-score for Class 1: The precision, recall, and F1-score for class 1 are all 0. This indicates the model's complete inability to distinguish instances of class 1 from class 0, resulting in zero true positives for stroke detection.

Therefore, it is evident that dealing with highly unbalanced data is crucial for developing effective classification models. Failing to address this imbalance leads to unrealistic training outcomes, where accuracy becomes an inadequate metric for evaluating model performance. Instead, metrics that consider the model's ability to correctly identify minority class instances, such as precision, recall, and F1-score, are more appropriate for assessing classification models in such scenarios.

The integration of machine learning (ML) in stroke prediction has the potential to greatly enhance Clinical Decision Support Systems (CDSS) by providing healthcare professionals with advanced tools for early diagnosis and risk assessment. ML models can analyze patient data in real-time, identifying high-risk individuals and recommending preventive measures, leading to improved patient outcomes and reduced strain on healthcare resources. By leveraging predictive analytics, CDSS can support clinicians in making more informed decisions, ensuring timely interventions, and optimizing patient care. For ML-based stroke prediction models to be successfully incorporated into CDSS, several critical aspects must be considered, including data integration, validation, user interface design, real-time alerting, and regulatory compliance[30,33].

Seamless data integration and interoperability with electronic health record (EHR) systems is essential for real-time patient monitoring. Standardized healthcare protocols such as HL7 and FHIR ensure that ML models can function within existing medical workflows without disrupting routine practice.

Another key factor is clinical validation and model reliability. To gain acceptance among healthcare professionals, ML models must undergo rigorous testing with diverse patient datasets across multiple institutions. Retrospective studies and clinical trials play a crucial role in assessing model performance and ensuring compliance with regulatory standards for medical AI applications. Furthermore, model transparency and interpretability are essential for building clinician trust, as doctors need to understand how predictions are made to confidently use these tools in decision-making.

Ensuring compliance with data protection laws such as GDPR and HIPAA is essential when integrating ML models into clinical practice[33]. Robust security measures must be implemented to protect patient confidentiality

and prevent unauthorized data access. Furthermore, ethical concerns such as bias mitigation and fairness in predictions must be addressed to ensure that stroke prediction models perform reliably across diverse patient populations, minimizing disparities in healthcare outcomes.

Conclusions

Various machine learning models were constructed to address the challenge of predicting stroke occurrence using a highly unbalanced dataset, including Decision Tree, Random Forest, K-Nearest Neighbors, Ada Boost, and Stacking. Among these models, the Random Forest Classifier emerged as the most effective. Employing the Grid Search method for hyperparameter optimization yielded the best results, with the Random Forest Classifier again proving the optimal choice. The best hyperparameters for the Random Forest Classifier were determined as n_estimators – 500, criterion – entropy, max_depth – 30. This configuration resulted in impressive performance metrics: Precision – 90%, Recall – 90%, F-1 Score – 90%. These findings hold significant implications for medical practitioners, as they can enhance the implementation of preventive measures, thereby improving the prognosis and outcomes for stroke patients. The practical conclusion is that when the data is unbalanced, it is worth solving this problem because the results obtained from the model training will not correspond to reality. Studies have shown that the accuracy metric is inaccurate when validating classification models. It is necessary to take into account the peculiarities of data pre-preparation.

Data availability

The data presented in this study from DataHack : Biggest Data hackathon platform for Data Scientists - https://github.com/nmelnyko/data_stroke/blob/main/data_stroke.csv.

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Author contributions

N. M., Y. P., S. S., M. F., S. F., and K. D.: methodology, data curation, formal analysis. N. M., Y. P., S. S., M. F., S. F., and K. D.: conceptualization, visualization. N. M., Y. P., S. S., M. F., S. F., and K. D.: formal analysis, writing. All authors have read and agreed to the published version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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